Some model tests

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Some numerical test about training and testing of cross section fitting using neural networks

I. INTRODUCTION

We use a neural network as universal interpolator to fit calculations of total cross sections (TCS) within two theoretical models, namely the Continuum-Distorted-Wave with Eikonal-Initial -State (CDW-EIS) and the Classical Trajectory MonteCarlo (CTMC) for ionization of hydrogen by bare ion impact. Here we consider bare projectiles with charges Z_P ranging between 1 and 9

II. METHODS

We use PyTorch framework to optimize the fitting of a multilayer perceptron (MLP) over a set of pre-calculated ionization TCS.

The architecture of the MLP has an input of the projectile impact energy, the charge of the projectile and the theory used

- Impact energy: we use decimal logarithm to flat the data in usable ranges ≈ 10
- Projectile charge: casted to float
- Theory: we one-hot encode the theory used (CTMC, CDW-EIS and semi-empirical formula for proton impact by Rudd et al.)

To understand the representation capability (fitting capability) we use 3 identical hidden layers. We explore the case of 32, 64 or 128 hidden units per layer. For non-linearity we use as activation functions infinite-differentiable functions, namely: the so used tanh and a gaussian activation.

The output layer represent the TCS. In the training stage we optimize the network against the decimal logarithm of the TCS in cm^2 to deal with numbers around 10.

III. RESULTS

A. Training process

In Fig. 1 we show the losses evolution during the training process.

- As the number hidden units increase, the lower loss reached.
- The gaussian activation function has better performance

In Fig. 3 we show the evolution during the training process for several global metrics that compute the error

- As the number hidden units increase, MAE and MRE seems to converge for gaussian activation.
- tanh present some overfitting.
- The gaussian activation function has slightly better performance

In Fig. **??** we show the evolution during the training process for several maximum of the former metrics to understand the deviations over each model

- This plot shows the power of gaussian activation.

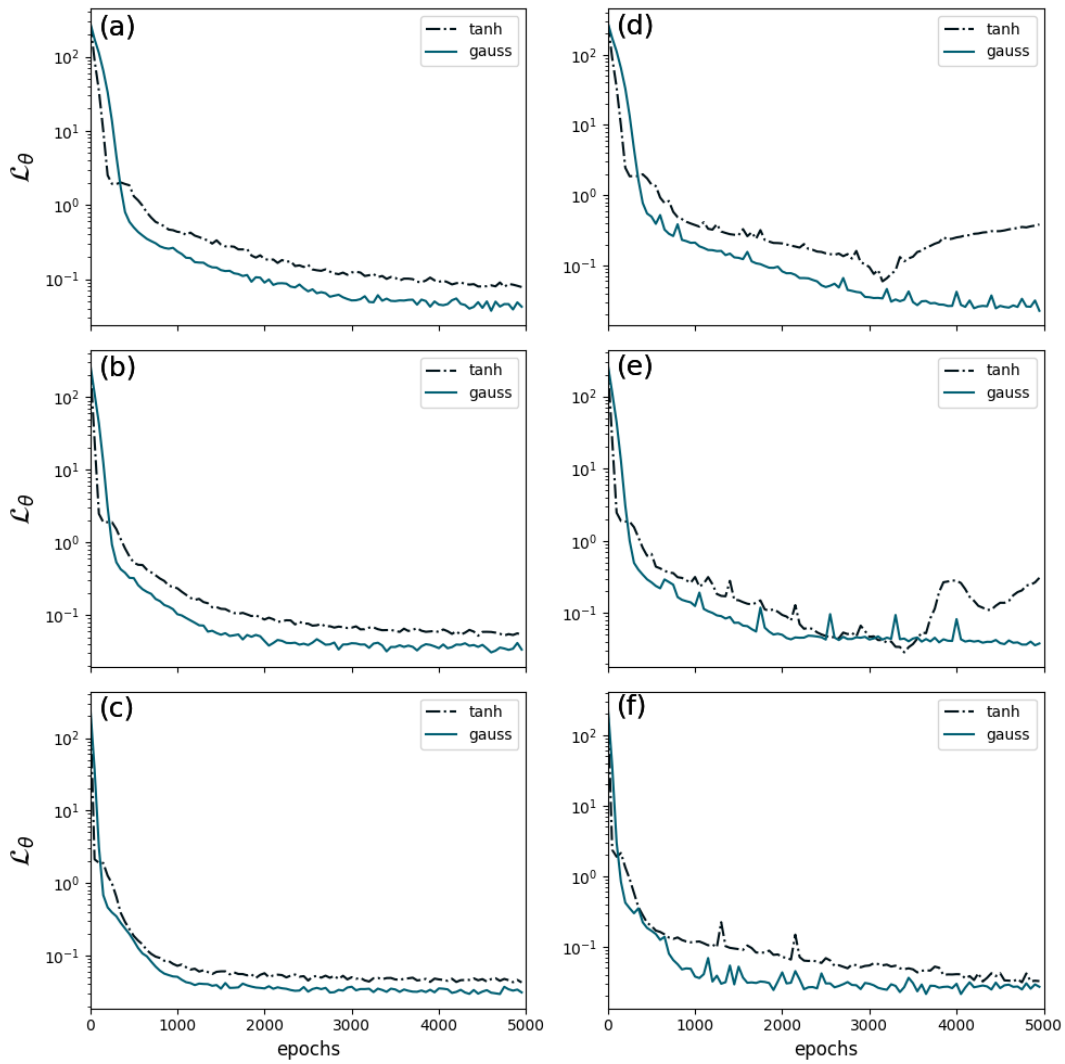

FIG. 1. Training losses over epochs for different network architectures.Panels (a-c) loss function for the training set. Panels(d-f) loss function for the validation set. Each row correspond to 32 (panels (a) and (d)), 64 (panels (b) and (e)) and 128 (panels (c) and (f)) hidden units per layer respectively.

B. Evaluation stage

We perform some evaluations over the whole TCS training set with the best models (128 hidden units) to understand the different errors in cm^2. Even when we have determined that gaussian activation has better performance, we show results for both activation types

- This plot shows the power of gaussian activation.

IV. DISCUSSION

- WE NEED TO DO THE EXPERIMENT WITH ZP HOLES AND ENERGY EXTRAPOLATIONS
- WE NEED TO DEFINE RANGES OF INPUT ENERGIES IN THE INPUT PIPELINE
- SEBASTIAN HAS TO PLOT THE SEMIEMPIRIC FIT!!!!
- MISSING THE BENCHMARKING AGAINST THE CUBIC INTERPOLATOR. ASK TO MICHELE IN DEEP.

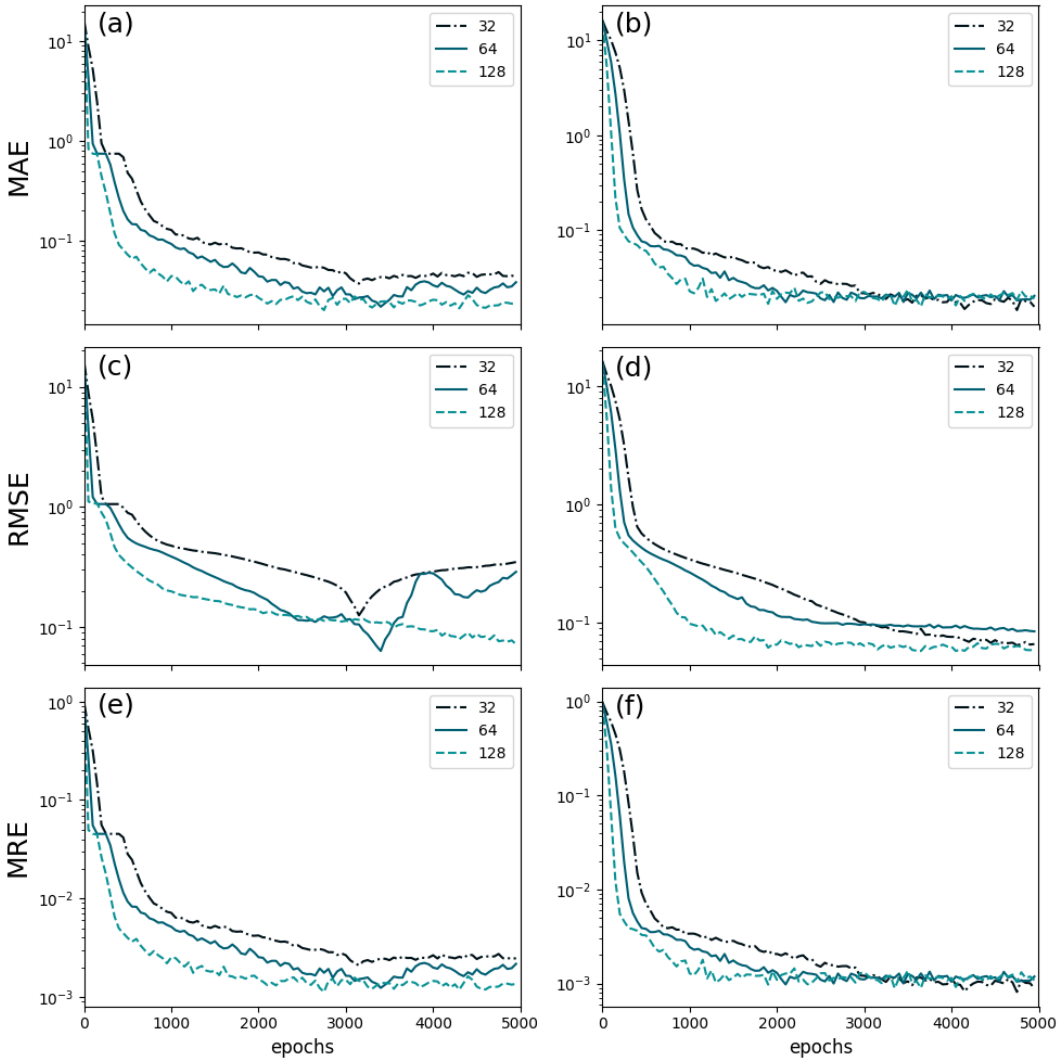

FIG. 2. Metrics over validation set for different network architectures. Left column (a, c, and e) metrics with tanh activation function. Right column (b-d-f) gaussian activation function. Each row correspond to a different metric over the $\log_{10}$ of the TCS.

V. CONCLUSION

Summarize your conclusions and future work.

[1] Author, "Title," *Journal*, vol. X, pp. XX-XX, Year.

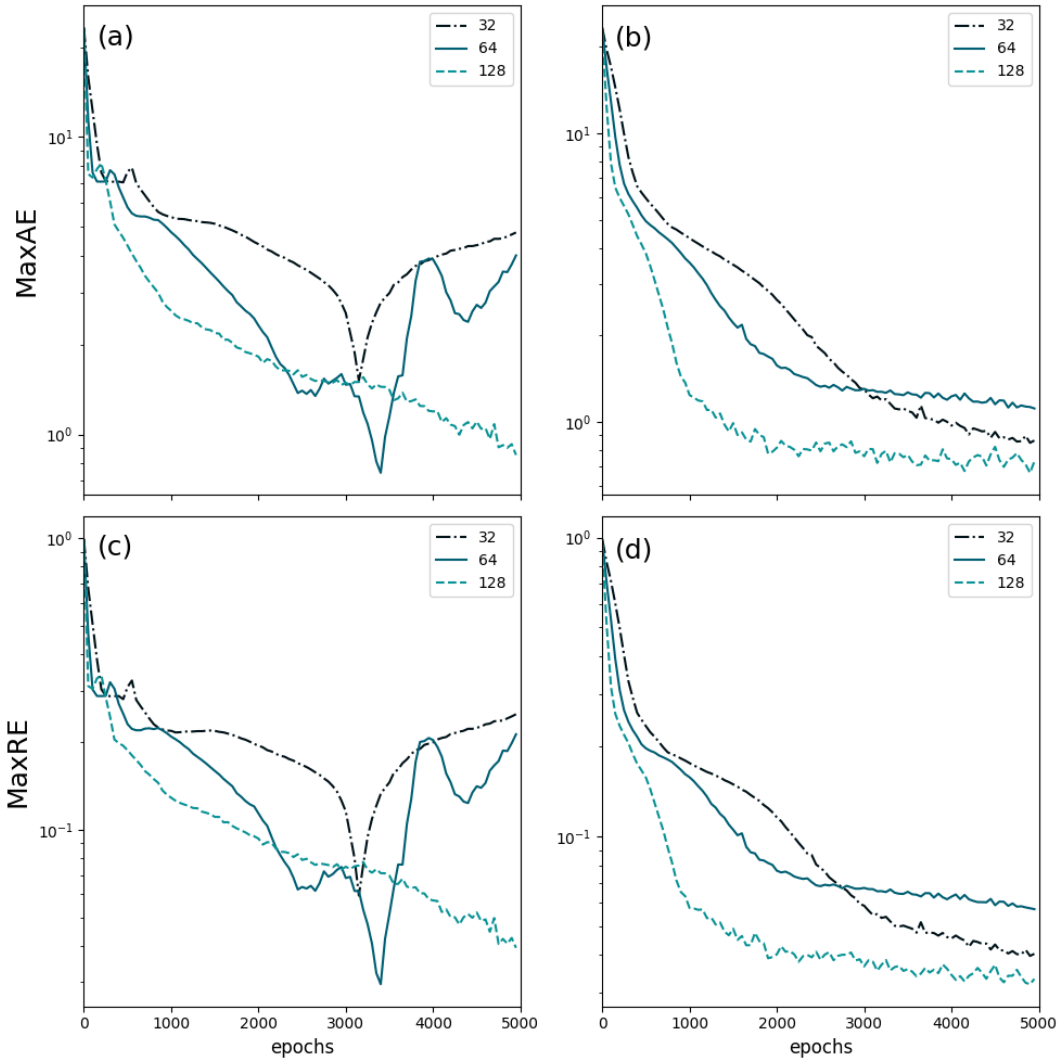

FIG. 3. Maximum values of Metrics over validation set for different network architectures. Left column (a, c) metrics with tanh activation function. Right column (b-d) gaussian activation function. Each row correspond to a different metric over the $\log_{10}$ of the TCS.

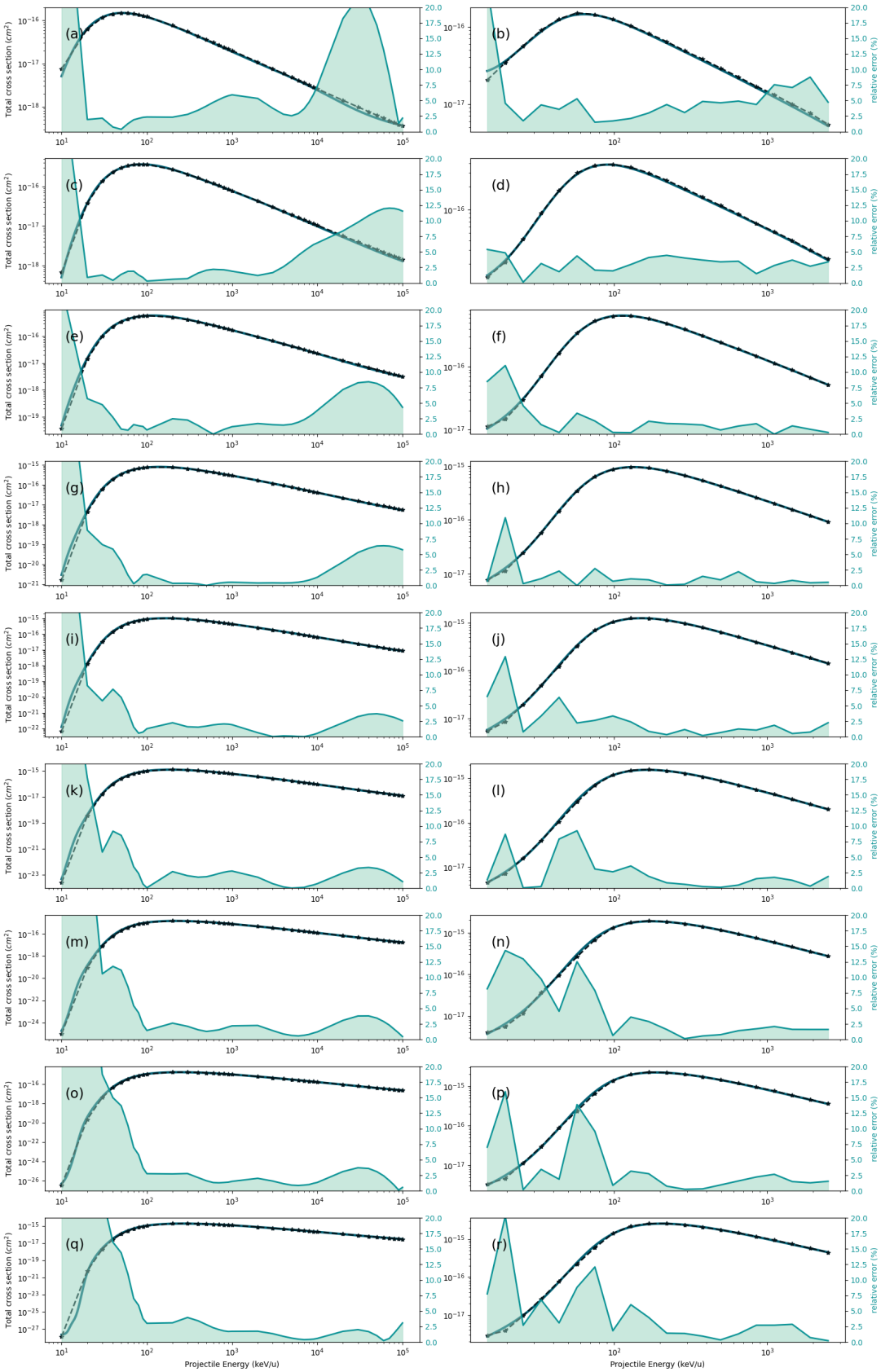

FIG. 4. Prediction of the MLP for CDW-EIS (left column) and CTMC (right column) as a function of the projectile impact energy. Each row correspont to a different projectile charge from $Z_P = 1$ to $Z_P = 9$. Activation function: tanh

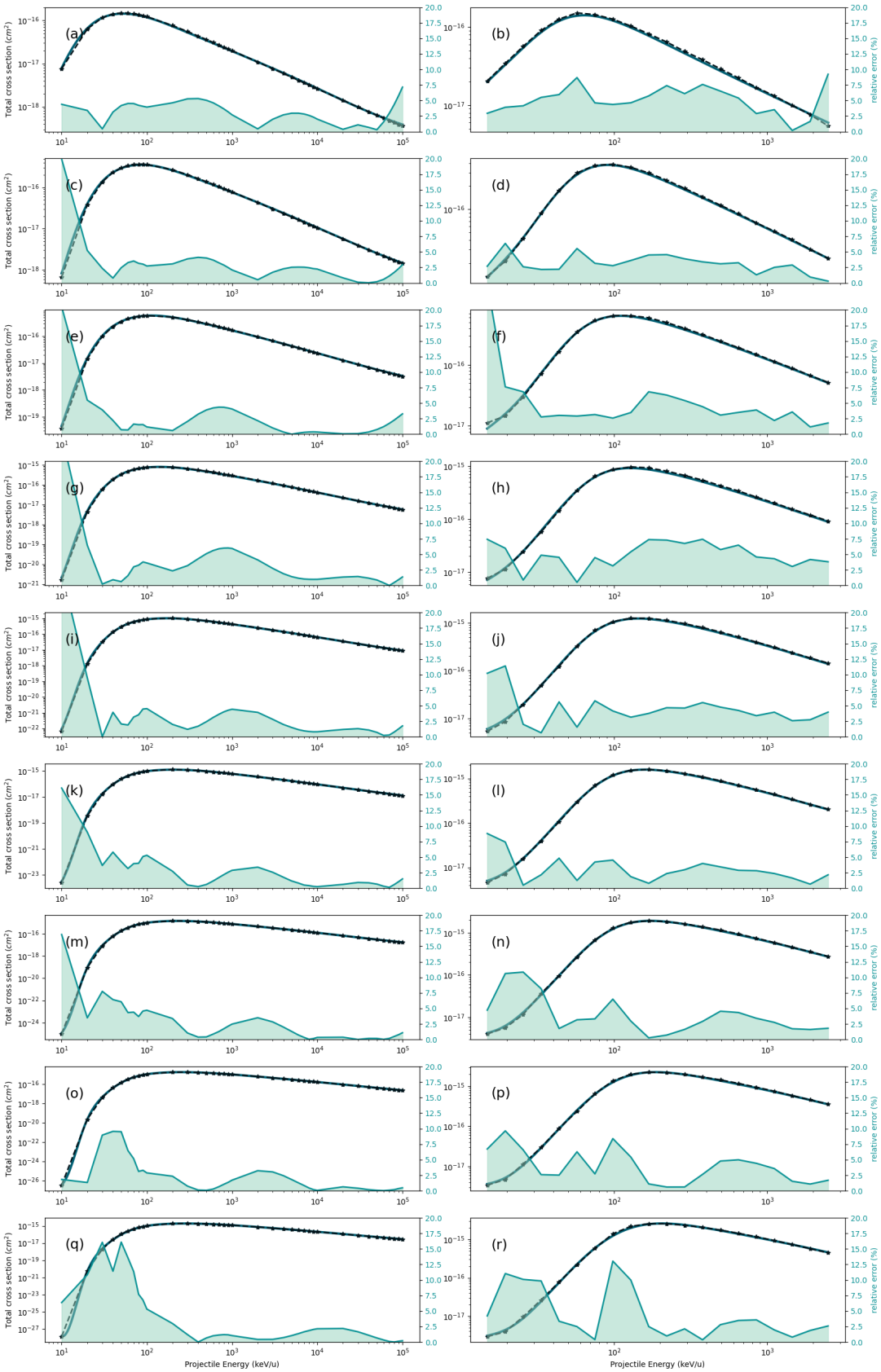

FIG. 5. Prediction of the MLP for CDW-EIS (left column) and CTMC (right column) as a function of the projectile impact energy. Each row correspont to a different projectile charge from $Z_P = 1$ to $Z_P = 9$. Activation function: gaussian.